

Satellite Orbit Simulator

Python-based satellite orbit propagation, visualization and closest-approach analysis

1. Project Objective

The Satellite Orbit Simulator is a Python software project designed to explore practical applications of orbital mechanics and scientific programming. The system takes Two-Line Element (TLE) data for satellites, propagates their orbits using the SGP4 model, generates time-series position data, visualizes orbital behaviour, and compares the separation of multiple satellites.

The project was developed as a self-directed engineering project with the aim of combining programming, numerical methods and space-engineering concepts in one reproducible workflow.

2. Input Data and TLE Processing

Satellite orbital information is supplied through TLE records stored in **data/sample_tles.txt**. The current sample contains the ISS (ZARYA), HUBBLE SPACE TELESCOPE and NOAA 15. The TLE loader reads each satellite name and its two orbital-element lines and stores them in a structured representation for later processing.

Basic validation is performed to ensure that the expected TLE structure is present before the data is passed to the propagation stage.

3. Orbit Propagation

The SGP4 (Simplified General Perturbations 4) model is used to propagate each satellite's state from its TLE. At each selected time, the program calculates a position vector in kilometres and a velocity vector in kilometres per second. Instead of analysing only one instant, the simulator generates a sequence of states at regular time intervals.

The resulting time series provides the numerical foundation for the visualisation and satellite-comparison stages of the project.

4. Orbit Visualisation

The simulator produces graphical representations of the propagated orbit. An XY trajectory plot shows the changing position of the satellite, while an altitude-versus-time plot illustrates the variation in orbital altitude during the simulated interval.

Output	Purpose
orbit_xy.png	Visualises the propagated orbital trajectory
altitude_vs_time.png	Shows altitude variation throughout the simulation
distance plots	Show separation between selected satellite pairs

5. Multi-Satellite Comparison

The analysis supports simultaneous propagation of multiple satellites. The ISS is used as the target satellite in the current example, with its propagated positions compared against the Hubble Space Telescope and NOAA 15.

For each pair, the Euclidean separation between the two position vectors is calculated at each common timestep. The smallest calculated separation is recorded together with the corresponding time.

6. Closest Approach and Risk Screening

A closest-approach search identifies the minimum sampled separation between the target and each comparison satellite. A simple distance threshold is then used to classify the result as either CLEAR or FLAGGED. In the current configuration the screening threshold is 5 km.

This threshold is deliberately presented as a simple screening mechanism. It is not intended to represent a professional collision-probability calculation or an operational conjunction assessment.

Comparison	Example minimum separation	Classification
ISS – Hubble	2433.31 km	CLEAR
ISS – NOAA 15	8435.90 km	CLEAR

These values are representative outputs from the current sample TLE dataset and propagation window. Results change when the input TLE epoch or simulation interval changes.

7. Data Export and Software Structure

Closest-approach results are exported to `closest_approaches.csv`. The source code is organised into separate modules for TLE loading, propagation, plotting, closest-approach calculation, comparison and data export. This separation makes individual components easier to test and modify.

8. Verification and Testing

Automated tests are implemented using Pytest. The current test suite contains four tests covering distance calculation, closest-approach calculation, TLE loading and SGP4 propagation. The complete suite currently passes with **4 passed**.

9. Limitations

- TLE data is not perfectly accurate and becomes less representative as it ages.
- SGP4 is a propagation model rather than a complete physical simulation of every perturbation.
- Closest approaches are evaluated at discrete time steps, so an event between samples could be missed.
- The 5 km risk threshold is a simplified screening rule rather than a collision probability.
- A professional conjunction assessment would require higher-quality orbital data, uncertainty/covariance information and more advanced analysis.

10. Future Development

Potential improvements include automatic retrieval of current TLE data, configurable simulation parameters, interactive 3D orbit visualisation, ground-track mapping, uncertainty and covariance modelling, more sophisticated conjunction assessment, and automated report generation.

11. Conclusion

The completed project demonstrates a complete numerical workflow from satellite TLE input through SGP4 propagation, visualisation, multi-satellite comparison, closest-approach detection, risk screening, CSV export and automated testing. It combines software engineering with a practical space-engineering problem and provides a foundation for future work involving more advanced orbital analysis.